

Genome-wide maps of histone modifications unwind in vivo chromatin states of the hair follicle lineage.

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Evidence abstract

Using mouse skin, where bountiful reservoirs of synchronized hair follicle stem cells (HF-SCs) fuel cycles of regeneration, we explore how adult SCs remodel chromatin in response to activating cues.

By profiling global mRNA and chromatin changes in quiescent and activated HF-SCs and their committed, transit-amplifying (TA) progeny, we show that polycomb-group (PcG)-mediated H3K27-trimethylation features prominently in HF-lineage progression by mechanisms distinct from embryonic-SCs.

In HF-SCs, PcG represses nonskin lineages and HF differentiation.

In TA progeny, nonskin regulators remain PcG-repressed, HF-SC regulators acquire H3K27me3-marks, and HF-lineage regulators lose them.

Interestingly, genes poised in embryonic stem cells, active in HF-SCs, and PcG-repressed in TA progeny encode not only key transcription factors, but also signaling regulators.

We document their importance in balancing HF-SC quiescence, underscoring the power of chromatin mapping in dissecting SC behavior.

Our findings explain how HF-SCs cycle through quiescent and activated states without losing stemness and define roles for PcG-mediated repression in governing a fate switch irreversibly.

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